A function $f : A \to B$ is called many-one function if two or more elements of A have same image in B.

In other words, a function $f : A \to B$ is called many-one if f it is not one–one.

Illustration 11

Let $A = \{1,2,3,4\}$ and $B = \{a,b,c\}$, $f = \{(1,a), (2,a), (3,b), (4,c)\}$

Then f is a function from A to B in which different elements 1 and 2 of A have the same image a in B. Hence f is a many – one function.

1.7.3 Onto function

In a mobile phone assume that there are 3 persons in the contact. If every person in the contact receives a call, then the function representing making calls will be onto. (Fig.1.25)

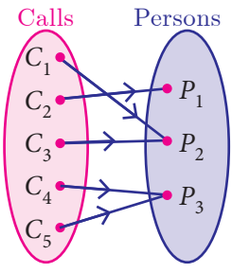


Fig. 1.25

A function $f : A \to B$ is said to be onto function if the range of f is equal to the co-domain of f.

In other words, every element in the co-domain B has a pre-image in the domain A.

An onto function is also called a **surjection**.

Note

If $f : A \to B$ is an onto function then, the range of $f = B$.

Illustration 12

Let $A = \{x, y, z\}, B = \{l, m, n\}$;

Range of $f = \{l, m, n\} = B$ (Fig.1.26)

Hence f is an onto function.

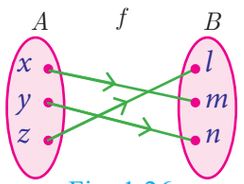


Fig. 1.26

1.7.4 Into function

In a home appliance showroom, the products television, air conditioner, washing machine and water heater were provided with 20% discount as new year sale offer. If the selection of the above products by the three customers C_1, C_2, C_3 is considered as a function then the following diagram (Fig.1.27) will represent an into function.

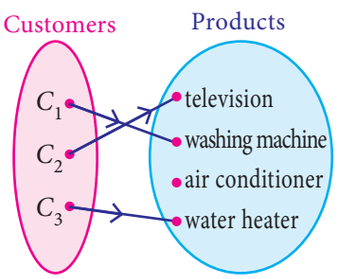


Fig. 1.27

During winter season customers usually do not prefer buying air conditioner. Here air conditioner is not chosen by any customer. This is an example of into function.

1 Relation and Functions.indd 18 15-12-2020 02:50:16 PM